



UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Academic Year 2020/2021 – First Year Examination – Semester I – 2021

SCS 1204 – Discrete Mathematics I

TWO (2) HOURS



Important Instructions to candidates:

1. The medium of instruction and questions is **English**.
2. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
3. Note that questions appear on both sides of the paper. If a page is not printed, please inform the supervisor immediately.
4. Write your index number on each and every page of the answer paper.
5. This paper has **4** questions and **04** pages.
6. Answer **ALL** questions. All questions carry equal marks (**25** marks).
7. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
8. **Non-Programmable/Programmable** calculators are **not allowed**.

1.
 - (a). Construct a truth table for the compound proposition $(p \vee \neg q) \rightarrow q$. [5 Marks]
 - (b). Using truth tables show that compound propositions $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$ are logically equivalent. [5 Marks]
 - (c). Let $E(x)$, $O(x)$, and $P(x)$ be three predicates over the set of all positive integers ($\mathbb{Z}^+$), where $E(x)$ denotes “ x is even”, $O(x)$ denotes “ x is odd”, and $P(x)$ denotes “ x is prime”. Express each of these statements in terms of $E(x)$, $O(x)$, $P(x)$, quantifiers, and logical connectives.
 - i. Every positive integer is even or odd.
 - ii. Not all primes are odd.
 - iii. If a positive integer is not odd, then it is even.
 - iv. The only even prime is 2.
 - v. Addition of two prime numbers need not be a prime number.
 - vi. Product of any two prime numbers is not a prime number. [9 Marks]
 - (d). Show that the following argument is valid:
 Alan will be at the park if Chandra is there and Bill is not there. Chandra will be at the park if it is a Friday or Saturday. If Bill is at the park, Denesh will be there. Denesh won't be at the park if it is a Friday. Today is a Friday. Therefore, Alan will be at the park. [6 Marks]
2.
 - (a). Consider the statement “No dogs are intelligent”.
 - i. Express this statement in predicate logic without using the existential quantifier.
 - ii. Express this statement in predicate logic without using the universal quantifier. [6 Marks]
 - (b). Prove that for all $x \in \mathbb{R}$, $|x + 3| - x > 2$. [4 Marks]
 - (c). Prove that for any two integers a, b ; if ab is even, then at least one of a and b is even. (Hint: use proof by contraposition). [5 Marks]
 - (d). Determine the truth value of each of the following statements if the domain of each variable consists of all real numbers:

- i. $\forall x \exists y [x^2 = y]$;
- ii. $\exists x \forall y [xy = 0]$;
- iii. $\forall x [x \neq 0 \rightarrow \exists y (xy = 1)]$;
- iv. $\forall x \exists y [x = y^2]$;
- v. $\forall x \exists y [x + y = 1]$.

[5 Marks]

(e). Use the mathematical induction principle to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n . [5 Marks]

3.

(a). Let $U = \{n \in \mathbb{Z} \mid 1 \leq n \leq 15\}$. Suppose that $A = \{n \in U \mid n \text{ is an even number}\}$, and $B = \{n \in U \mid n \text{ is a prime number}\}$.

i. Represent sets U, A, B in a Venn diagram.

ii. Find the following sets:

- $A \cap B$;
- $A \cup B$;
- $A - B$;
- A^c ;
- $A \oplus B$.

[7 Marks]

(b). Let A, B , and C be three sets. With the usual notation, prove the following set identities:

i. $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$;

ii. $(A - B) \cap (C - B) = (A \cap C) - B$.

[7 Marks]

(c). Give an explicit formula for a function from the set of natural numbers ($\mathbb{N}$) to itself that is

- i. one-to-one, but not onto;
- ii. onto, but not one-to-one; and
- iii. not one-to-one and not onto.

[6 Marks]

(d). Let f be a function from a set A to a set B . Let S and T be any two subsets of B . The inverse image of S under f is the subset of A , denoted by $f^{-1}(S)$, and defined as $f^{-1}(S) = \{x \in A \mid f(x) \in S\}$. Show that

- i. $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T);$
- ii. $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$

[5 Marks]

4.

(a). Let R be the relation on the set $\{1, 2, 3, 4, 5\}$ containing the ordered pairs $(1, 3), (2, 4), (3, 1), (3, 5), (4, 3), (5, 1), (5, 2),$ and $(5, 4).$

- i. Is R a transitive relation? Justify your answer.
- ii. Represent the relation R with a matrix.
- iii. Find the transitive closure of R .

[12 Marks]

(b). Suppose that $A = \{3, 5, 9, 15, 24, 45\}$ and R is the partial order relation defined on A by $(x, y) \in R$ if and only if x divides y .

- i. Draw the Hasse diagram for the relation R .
- ii. Find all minimal elements of A .
- iii. Find all maximal elements of A .
- iv. Find the least element of A (if exists).
- v. Find the greatest element of A (if exists).
- vi. Find all upper bounds of $\{3, 5\}$.
- vii. Find the least upper bound of $\{3, 5\}$, if it exists.
- viii. Find all lower bounds of $\{15, 45\}$.
- ix. Find the greatest lower bound of $\{15, 45\}$, if it exists.

[13 Marks]